

2D Navier–Stokes recurrent adversarial attack

(mode paths: $c_1, \dots, c_9, c_{19}$; recurrent prediction 10–19)

→ $\mathcal{G}$ forward

→ $\mathcal{F}_\theta$ forward

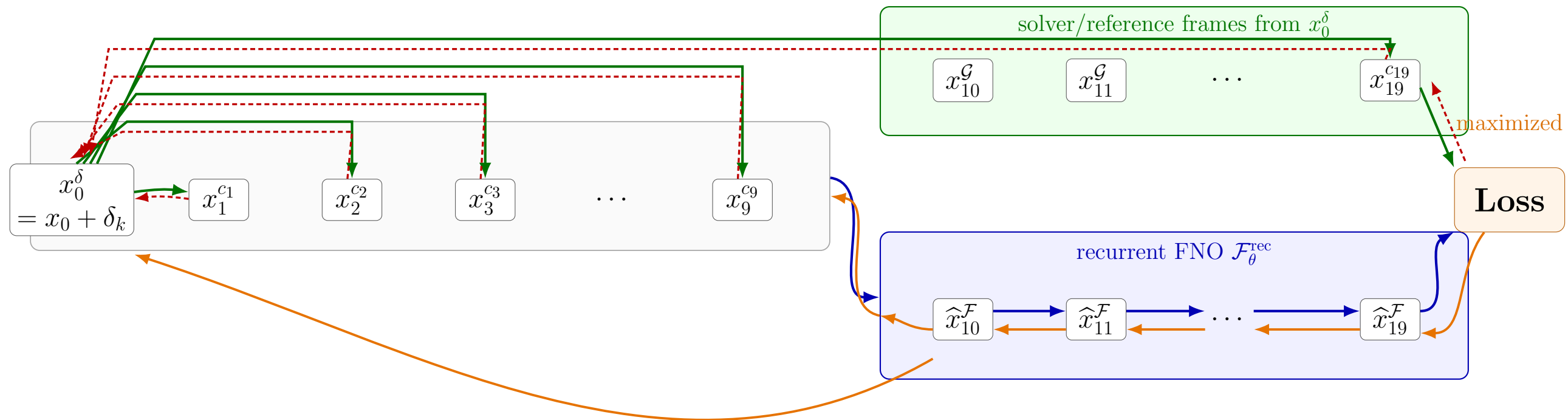
→ $\mathcal{F}_\theta$ backward

→ $\mathcal{G}$ backward (W only; A/D : no backprop)

$$x_0^\delta = x_0 + \delta_k, \quad \text{mode_spec} = (c_1, \dots, c_9, c_{19}), \quad c_j \in \{A, D, W\}$$

$$[\hat{x}_{10}^{\mathcal{F}}, \dots, \hat{x}_{19}^{\mathcal{F}}] = \mathcal{F}_\theta^{\text{rec}}([x_0^\delta, x_1^{c_1}, \dots, x_9^{c_9}]), \quad \mathcal{L}_{\text{adv}}(\delta_k) = \|\hat{x}_{19}^{\mathcal{F}} - x_{19}^{c_{19}}\|_2^2$$

A : dictionary/approx.; D : detached solver; W : solver backprop
 $c_1, \dots, c_9$: input window, c_{19} : loss target



$$\nabla_{\delta_k} \mathcal{L}_{\text{adv}} = D\mathcal{G}_{0:9}^* D(\mathcal{F}_\theta^{\text{rec}})^* \nabla_{\hat{x}_{19}} \ell + D\mathcal{G}_{19}^* \nabla_{x_{19}^{\mathcal{G}}} \ell$$